

Income Insurance and the Equilibrium Term-Structure of Equity

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1 Motivation & Aim

2 Model Intuition

3 Empirical Support

4 Model

5 Main Results

6 Model Extension

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2 Model Intuition

3 Empirical Support

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6 Model Extension

Motivation and Aim

- Leading asset pricing models provide a **long-term** explanation of standard puzzles.
- Recent empirical research documents some important features of dividends and equity:
 - **downward-sloping** term structures of dividends volatility (Belo et al. 2015 JF),
 - positive premium for **short cash-flows duration** firms (Dechow et al. 2004 RAS, Weber 2016 wp),
 - **downward-sloping** term structures of equity volatility and premia (van Binsbergen Kojien et al. 2012 AER, 2013 JFE, 2016 JFE),
 - **inconsistent** with standard models,
 - question the **nature** of standard asset pricing facts.
- The term-structures of both cash-flows and equity returns provide **information** about how prices are determined in equilibrium –disregarded in standard models.
- A term-structure perspective offers additional **testable implications** to asset pricing frameworks and can help to understand the macroeconomic determinants of asset prices.

Motivation and Aim

I argue that labor relations, namely a mechanism of **income insurance from shareholders to workers**, are the main determinant of the term-structure properties of both dividends and equity.

Why are labor relations important to asset pricing?

Intuition from standard models:

$$D \approx Y - W - I - B$$

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$$\underbrace{\frac{D}{Y}}_{\text{counter-cyclical}} \approx 1 - \underbrace{\frac{W}{Y}}_{\text{constant}} - \underbrace{\frac{I}{Y}}_{\text{pro-cyclical}} - \underbrace{\frac{B}{Y}}_{\text{small}}$$

- Dividends cannot command a premium!
- Failure of RBC models to match asset prices.

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Actual data:

- labor-share is **counter-cyclical**,
 - labor- and dividend-shares are strongly **negatively** correlated,
- distributional risk and leverage effect on dividends!

Motivation and Aim

Labor rigidity allows for labor-share imperfectly correlated with output.

New contribution: labor relations alter not only the **cyclicality** of dividends but also the **timing** of dividend risk.

Aims

Motivation and Aim

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Aims

1. Recognize heterogeneity in the term-structure of risk of macroeconomic variables
2. Recognize labor relations as the main economic channel behind the timing of risk
3. Reconcile the term-structure of fundamentals and equity with standard asset pricing facts in a closed-form general equilibrium model
4. Investigate recursive preferences in presence of permanent and transitory shocks
5. Exploit term-structures information to infer about persistence of latent factors
6. Additional testable implications: term-structure of interest rates, counter-cyclical Sharpe ratio, predictability, cross-sectional value premium

Motivation and Aim

Some related literature

- Income insurance within the firm:

Knight (1921), Baily (1974 REStud), Azariadis (1978 JET), Gamber (1988 JLE), Boldrin Horvath (1995 JPE), Gomme Greenwood (1995 JEDC), Guiso Pistaferri Schivardi (2005 JPE), Shimer (2005 AER), Menzio (2005), Gertler Trigari (2009 JPE), Ríos-Rull Santaaulèlia-Llopis (2010 JME), Han Maug Schneider (2013), Ellul Pagano Schivardi (2016), ...

- Labor markets and asset prices:

Danthine Donaldson (1992 EER, 2002 REStud), Santos Veronesi (2006 RFS), Danthine Donaldson Siconolfi (2006), Gourio (2008), Lochstoer Bhamra (2009), Kuehn Petrosky-Nadeau Zhang (2012), Greenwald Lettau Ludvigson (2014), Donangelo (2014 JF), Favilukis Lin (2015 RFS), Lettau Ludvigson Ma (2016), ...

- Models for the term-structures of dividends and equity:

Lettau Watcher (2007 JF, 2011 JFE), Ai Croce Diercks Li (2013), Andries Eisenbach Schmalz (2014), Belo Collin-Dufresne Goldstein (2015 JF), Croce Lettau Ludvigson (2015 RFS), Hasler Marfè (2016 JFE), Marfè (2016 RF), ...

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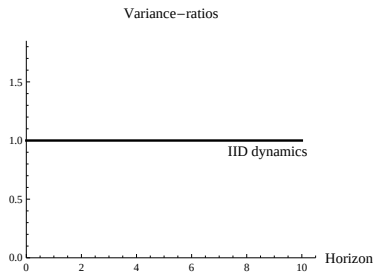
6 Model Extension

Intuition

- Consider the following dynamics:

$$d \log X_t = \mu dt + \sigma_x dB_{x,t}$$

- Variance ratios of growth rates:

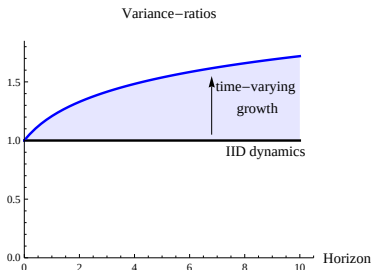


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$$\begin{aligned}d\log X_t &= \mu_t dt + \sigma_x dB_{x,t} \\ d\mu_t &= \lambda_\mu(\bar{\mu} - \mu_t)dt + \sigma_\mu dB_{\mu,t}\end{aligned}$$

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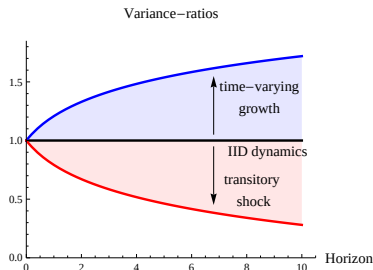
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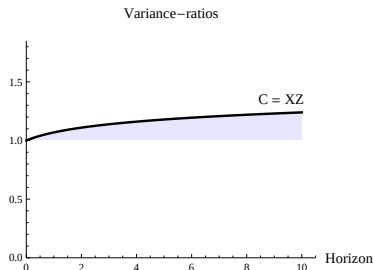
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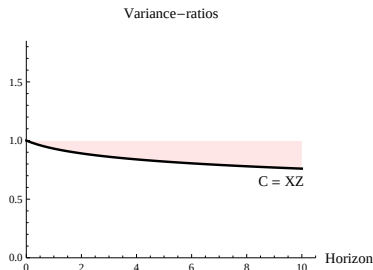
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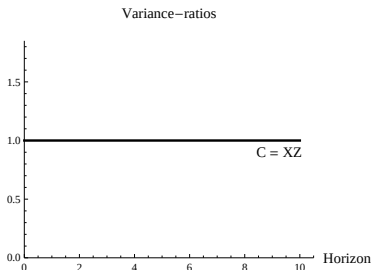
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- Assume:

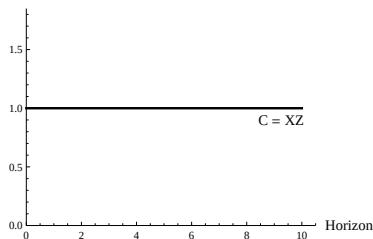
$$W_t = X_t g(Z_t) \quad \text{and} \quad D_t = X_t h(Z_t)$$

with

$$g'(Z) > 0, \quad g''(Z) < 0 \quad \text{and} \quad h(Z) = Z - g(Z) \Rightarrow W_t + D_t = C_t$$

- Variance ratios of growth rates:

Variance-ratios



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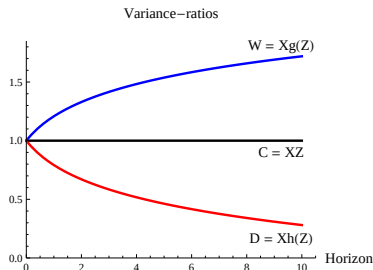
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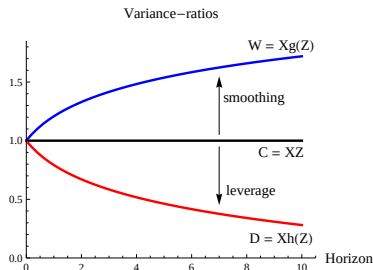
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- Variance-ratios of growth rates:

Economic interpretation:

Risk sharing of
transitory risk



Ciclycality effect:

$$W/C'(z) < 0 \quad D/C'(z) > 0$$

Term-structure effect:

$$\sigma'_W(\tau) > 0 \quad \sigma'_D(\tau) < 0$$

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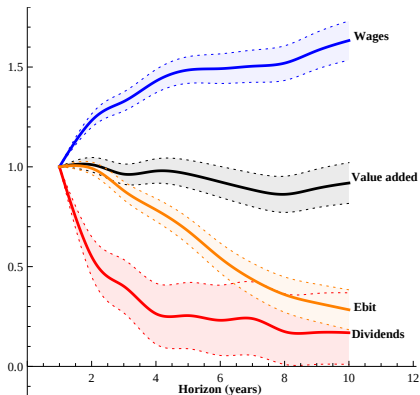
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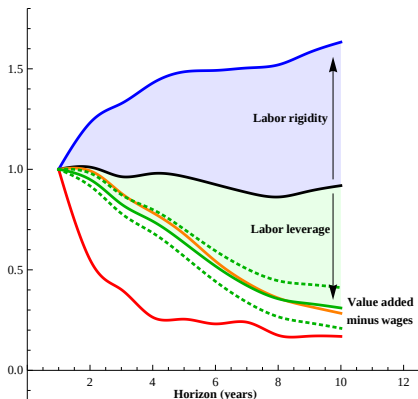
Empirical Support

- Variance-ratios (VR's) tell us how risk evolves with the horizon.
- The timing of macroeconomic risk is heterogeneous:



Empirical Support

- Leading models overestimate long-horizon dividend risk.
- Whence the negative slope of dividend risk?
- Simple idea: look at the remainder of output minus wages



The Term-Structure Effect of Income Insurance

- Danthine and Donaldson (EER 1992, REStud 2002) show that risk sharing within the firm leads to operating leverage, risky dividends and high equity compensations.
- Some empirical findings:
 1. labor share is counter-cyclical [Ríos-Rull Santaeulèlia-Llopis JME 2010]
 2. output, wages, dividends are co-integrated [Lettau Ludvigson JFE 2005]
 3. macro variables are subject to permanent and transitory shocks [Lettau Ludvigson NBER Macro 2014]
 4. income insurance does not concern permanent shocks [Gamber 1988 JLE, Guiso Pistaferri Schivardi JPE 2005, Menzio 2005, Ellul Pagano Schivardi 2016]
- Beyond the cyclical effect of income insurance, also a term-structure effect takes place.
- Labor rigidity leads to upward-sloping wage risk and downward-sloping profit/dividend risk.
- Implications for equity and bonds.

The Term-Structure Effect of Income Insurance

- Income insurance implies that the labor-share (W/Y) moves:
 - **positively** with the VR's of wages,
 - **negatively** with the VR's of dividends and
 - **positively** with the distance between the VR's of wages and dividends.
- Time-variation is consistent with income insurance mechanism:

$$VR_W|_{t,\tau} = a + b_w \frac{W}{Y}_t + b_y VR_Y|_{t,\tau} + \epsilon_t,$$

$$VR_D|_{t,\tau} = a + b_d \frac{W}{Y}_t + b_y VR_Y|_{t,\tau} + \epsilon_t,$$

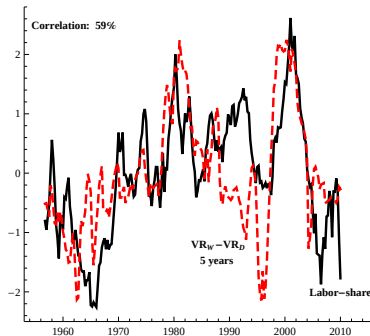
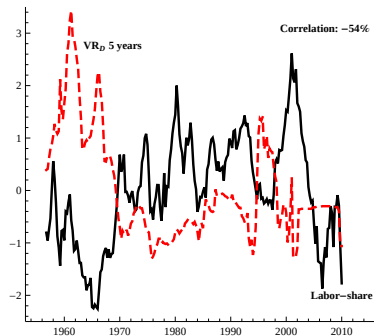
$$VR_W - VR_D|_{t,\tau} = a + b_{wd} \frac{W}{Y}_t + b_y VR_Y|_{t,\tau} + \epsilon_t,$$

τ	2	3	4	5	6	7
b_w	1.38	2.97	4.55*	6.51**	8.86**	11.43**
NW t-stat	(1.26)	(1.58)	(1.78)	(1.98)	(2.06)	(2.14)
adj-R ²	0.42	0.43	0.48	0.51	0.51	0.50

τ	2	3	4	5	6	7
b_d	-4.97***	-6.11***	-6.08***	-5.34***	-5.55***	-5.55***
NW t-stat	(-4.39)	(-5.11)	(-5.41)	(-4.67)	(-4.53)	(-4.21)
adj-R ²	0.32	0.31	0.31	0.29	0.33	0.34

τ	2	3	4	5	6	7
b_{wd}	6.35***	9.08***	10.63***	11.85***	14.41***	16.98***
NW t-stat	(5.50)	(5.97)	(4.55)	(3.74)	(3.45)	(3.17)
adj-R ²	0.42	0.53	0.55	0.57	0.58	0.56

The Term-Structure Effect of Income Insurance



- The slope of dividend risk decreases with the labor-share.
- Labor-share fluctuations shape the time-variation of the gap between the slopes of wage risk and dividend risk.

The Term-Structure Effect of Income Insurance

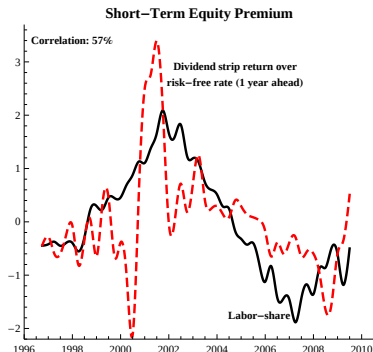
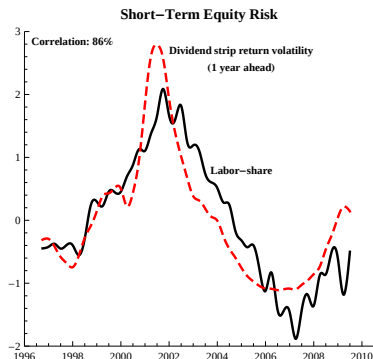
- How does income insurance affect dividend strip returns?

$$\begin{aligned} vol_{t+\Delta}^{\text{strip}} &= a + b_{\text{risk}} \frac{W}{Y}_t + \epsilon_t, \\ r_{t+\Delta}^{\text{strip}} - r_{t+\Delta}^{\text{risk-free}} &= a + b_{\text{premium}} \frac{W}{Y}_t + \epsilon_t, \\ r_{t+\Delta}^{\text{strip}} - r_{t+\Delta}^{\text{market}} &= a + b_{\text{slope}} \frac{W}{Y}_t + \epsilon_t, \end{aligned}$$

	Risk			Premium			Slope		
	1-quarter ahead	1-year ahead	2-years ahead	1-quarter ahead	1-year ahead	2-years ahead	1-quarter ahead	1-year ahead	2-years ahead
W/Y _t	1.10***	1.04***	0.88***	0.43*	0.39***	0.37***	0.58*	0.53**	0.43**
NW t-stat	(5.88)	(6.03)	(5.37)	(1.99)	(3.02)	(6.26)	(1.98)	(2.58)	(2.43)
adj-R ²	0.72	0.74	0.62	0.08	0.31	0.64	0.10	0.26	0.28

- Labor-share variation captures short-term equity risk and such a risk is compensated by markets.
- Bivariate and multivariate regressions: results are robust to Fama & French factors, market volatility, short-rate, price-earnings ratio, cay, financial leverage, investment share of income.

The Term-Structure Effect of Income Insurance



- Almost perfect one-to-one mapping between labor-share and dividend strip volatility.
- Markets compensate short-term equity risk induced by labor-share fluctuations.
- Labor-share half-life is ≈ 3.5 years ($\neq$ slow-frequency movements in Greenwald Lettau Ludvigson (2014)).

The Term-Structure Effect of Income Insurance

- Most of transitory risk of dividends is orthogonal to labor ($D/Y^\perp$): it **could** explain the slope of dividend risk and the properties of dividend strip returns. But it **does not**.

Dividend Variance Ratios

τ	2	3	4	5	6	7
W/Y	-4.97***	-6.11***	-6.08***	-5.34***	-5.55***	-5.55***
NW t-stat	(-4.42)	(-4.97)	(-5.23)	(-4.64)	(-4.58)	(-4.30)
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adj-R ²	0.34	0.32	0.32	0.28	0.33	0.34

Equity Risk, Premium and Slope

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$D/Y_t^\perp$	0.12	0.14	0.04	-0.02	0.02	-0.06**	0.32	0.39	0.24
NW t-stat	(1.15)	(1.28)	(0.31)	(-0.15)	(0.20)	(-2.45)	(-1.12)	(1.45)	(1.32)
adj-R ²	0.72	0.74	0.62	0.06	0.30	0.64	0.10	0.30	0.30

- Income insurance implies that both labor-share and expected dividend growth decrease with the transitory component of output $\Rightarrow$ **dividend growth predictability**.

Cumulative dividend growth

τ	1	5	10	15
W/Y	-0.12	2.69**	4.57***	6.93***
NW t-stat	(-0.24)	(2.54)	(2.76)	(4.08)
HH t-stat	(-0.49)	(3.33)	(2.37)	(3.64)
adj-R ²	0.00	0.10	0.24	0.37

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Economy and Income Insurance

Agents:

- Workers (w) and shareholders (s) feature stochastic differential utility.
- They have to share resources from a representative firm.
- Limited market participation: workers have no access to financial markets. (Berk Walden (RAPS 2013))

Representative firm:

- No investment, full employment,
- Operational cash-flows process, $C = XZ$, has dynamics:

$$\begin{aligned}d \log X_t &= dx_t = \mu_t dt + \sigma_x dB_{x,t}, \\d \mu_t &= \lambda_\mu (\bar{\mu} - \mu_t) dt + \sigma_\mu dB_{\mu,t}, \\d \log Z_t &= dz_t = \lambda_z (\bar{z} - z_t) dt + \sigma_z dB_{z,t}.\end{aligned}$$

- Subcases: $C = X$ long-run risk literature, $C = Z$ real business cycle literature,
- $C = XZ$ has flexible term-structures:
 - μ_t leads to upward sloping risk,
 - z_t leads to downward sloping risk.

Economy and Income Insurance

Income insurance:

- labor claim on output is partially fixed in advance: shareholders bear most of short-run aggregate uncertainty and, in exchange of income insurance, workers provide flexibility in labor supply.
- Income insurance is modelled by a contract rule:

$$\mathcal{C} = \{(W, D) : W = Xg(Z) \quad D = X(Z - g(Z)) \quad Y = W + D\}$$

- Smoothing function: $g(Z) = \alpha Z^{1-\phi}$, $\phi \in (0, 1)$.
- Interpretation of ϕ as the strength of income insurance resulting from bargaining negotiations and labor market frictions.
- Theoretical micro-foundation:
 - implicit contracts (Gamber (1988 JLE), Danthine Donaldson (1992 EER, 2002 REStud),
 - search frictions (Menzio (2005)).
- Simple reduced-form model guarantees analytical solutions and equilibrium stationarity.
- Walrasian benchmark:

$$\phi = 0 \Rightarrow W = \alpha Y, D = (1 - \alpha)Y.$$

Economy and Income Insurance

- Income insurance leads to counter-cyclical labor-share:

$$\omega(z_t) = \alpha Z_t^{-\phi}.$$

- Instantaneous risk: smoothing on wages and operating leverage on dividends:

$$(1 - \phi)\sigma_z = \sigma_z^W(t) < \sigma_z < \sigma_z^D(t) = \sigma_z \left(1 + \phi \frac{\omega(z_t)}{1 - \omega(z_t)} \right).$$

- The VR's of dividends decrease with the labor-share $\omega(z_t)$:

$$VR_D(\tau) = \frac{\sigma_D^2(t, \tau)}{\sigma_D^2(t, 1)}, \quad \text{with} \quad \sigma_D^2(t, \tau) = \frac{1}{\tau} \log \frac{\mathbb{E}_t[D_{t+\tau}^2]}{\mathbb{E}_t[D_{t+\tau}]^2}.$$

- The slope of wage and dividend risk increase and decrease with income insurance:

$$\frac{\partial^2}{\partial \phi \partial \tau} \sigma_W(t, \tau) > 0 \qquad \frac{\partial^2}{\partial \phi \partial \tau} \sigma_D(t, \tau) < 0.$$

- Rationale for downward-sloping dividend risk.

≠ bounded rationality in Croce Lettau Ludvigson (RFS 2015).

Equilibrium and State-Price Density

- I derive asset pricing implications under a log-linearized dynamics for dividends.
- State-price density satisfies (Duffie Epstein (ECMA 1992)):

$$d\xi_{0,t} = \xi_{0,t} (-r(t)dt - \theta_x dB_{x,t} - \theta_\mu dB_{\mu,t} - \theta_z dB_{z,t})$$

with

$$\theta_x = \gamma \sigma_x$$

→ price of instantaneous risk due to X_t ,

$$\theta_\mu = \frac{\gamma - 1/\psi}{e^{cq} + \lambda_\mu} \sigma_\mu$$

→ price of intertemporal risk due to long-run growth μ_t (LRR),

$$\theta_z = \frac{\gamma\psi e^{cq} + \lambda_z}{(e^{cq} + \lambda_z)\psi} \underbrace{\left(1 + \phi \frac{\alpha}{1 - \alpha}\right)}_{\text{leverage}} \sigma_z = \left(\gamma - \frac{\lambda_z(\gamma - 1/\psi)}{e^{cq} + \lambda_z}\right) dz \sigma_z$$

→ price of instantaneous and intertemporal risks due to z_t (SRR).

- Main implication: distributional risk due to income insurance is priced.
 ≠ partial equilibrium models by Lettau Wachter (JF 2007) and Belo et al. (JF 2015).

Asset Prices

Dividend strips price is given by:

$$P_{t,\tau} = X_t e^{A_0(\tau) + A_\mu(\tau)\mu_t + A_z(\tau)z_t}$$

The term-structures of equity premia and volatility are given by

$$(\mu_P - r)(t, \tau) = \gamma \sigma_x^2 + \frac{(1 - e^{-\lambda_\mu \tau})(1 - 1/\psi)(\gamma - 1/\psi)}{\lambda_\mu (e^{cq} + \lambda_\mu)} \sigma_\mu^2 + \frac{(\frac{1}{\psi}(1 - e^{-\lambda_z \tau}) + e^{-\lambda_z \tau})(\lambda_z + \gamma e^{cq})}{\psi(e^{cq} + \lambda_z)} d_z^2 \sigma_z^2$$

$$\sigma_P(t, \tau) = \sqrt{\sigma_x^2 + \frac{e^{-2\lambda_\mu \tau}(e^{\lambda_\mu \tau} - 1)^2(\psi - 1)^2}{\lambda_\mu^2 \psi^2} \sigma_\mu^2 + \frac{e^{-2\lambda_z \tau}(\psi + e^{\lambda_z \tau} - 1)^2}{\psi^2} d_z^2 \sigma_z^2}$$

The sign of the premium due to variation in long-run growth (LRR) depends on preferences $((1 - 1/\psi)(\gamma - 1/\psi))$, and increases with ψ if $\gamma > 1, \psi > 1$.

Whereas, the premium due to transitory shocks (SRR) is always positive but decreases with ψ .

Premia and volatility increase with the degree of income insurance ϕ .

Asset Prices

The slope of the term structure of equity premia is given by

$$\begin{aligned}\frac{\partial}{\partial \tau}(\mu_P - r)(t, \tau) &= (\psi - 1) \left(\frac{e^{-\lambda_\mu \tau} (\gamma - 1/\psi) \psi \sigma_\mu^2}{e^{cq} + \lambda_\mu} - \frac{e^{-\lambda_z \tau} (\gamma \psi e^{cq} + \lambda_z) \lambda_z d_z^2 \sigma_z^2}{e^{cq} + \lambda_z} \right) \\ &= \text{intertemporal substitution} \times \left(\text{resolution of uncertainty} \times \text{LRR} - \text{income insurance} \times \text{SRR} \right)\end{aligned}$$

Variation in long-run growth (LRR) commands premia increasing with the horizon if $\gamma > \psi^{-1}$, $\psi > 1$.

Transitory shocks (SRR) command premia decreasing with the horizon if $\psi > 1$, despite risk preferences.

The degree of income insurance ϕ enhances the downward sloping effect due to transitory shocks.

1 Motivation & Aim

2 Model Intuition

3 Empirical Support

4 Model

5 Main Results

6 Model Extension

Calibration, Cash-Flows and Income Insurance

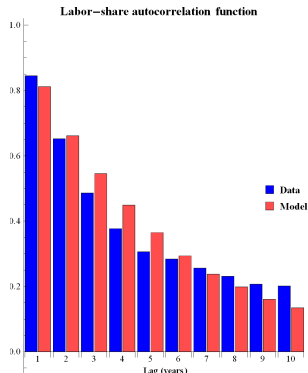
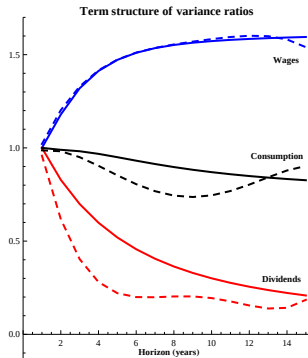
- I exploit analytical solutions to set the 8 cash-flows parameters.
- Peculiarity: I use information from the timing of macroeconomic risk:

long-run growth	$m_1(\theta) = \frac{ g_C^{\text{empirical}} - g_C }{g_C^{\text{empirical}}},$
short-run consumption vol	$m_2(\theta) = \frac{ \sigma_C(1)^{\text{empirical}} - \sigma_C(1) }{\sigma_C(1)^{\text{empirical}}},$
short-run dividends vol	$m_3(\theta) = \frac{ \sigma_D(1)^{\text{empirical}} - \sigma_D(1) }{\sigma_D(1)^{\text{empirical}}},$
dividend-share	$m_4(\theta) = \frac{ (1-\omega)^{\text{empirical}} - (1-\omega) }{(1-\omega)^{\text{empirical}}},$
dividend-share vol	$m_5(\theta) = \frac{ \sigma_{1-\omega}^{\text{empirical}} - \sigma_{1-\omega} }{\sigma_{1-\omega}^{\text{empirical}}},$
timing of consumption risk	$m_6(\theta) = \sum_{\tau=2}^{15} \frac{ VR_C^{\text{empirical}}(\tau) - VR_C(\tau) }{VR_C^{\text{empirical}}(\tau)},$
timing of wage risk	$m_7(\theta) = \sum_{\tau=2}^{15} \frac{ VR_W^{\text{empirical}}(\tau) - VR_W(\tau) }{VR_W^{\text{empirical}}(\tau)},$
timing of dividend risk	$m_8(\theta) = \sum_{\tau=2}^{15} \frac{ VR_D^{\text{empirical}}(\tau) - VR_D(\tau) }{VR_D^{\text{empirical}}(\tau)}.$

- Parameter vector minimizes the average relative error.

Calibration, Cash-Flows and Income Insurance

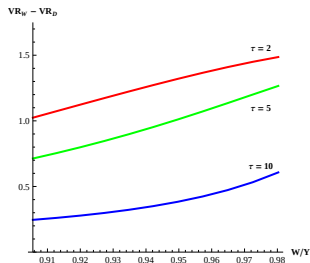
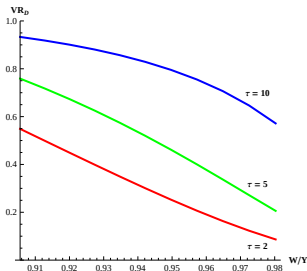
- Model captures the flat, upward-sloping and downward-sloping variance ratios of consumption, wages and dividends risk.



- VR's provide information about latent factors: model matches the labor-share autocorrelation function.

Calibration, Cash-Flows and Income Insurance

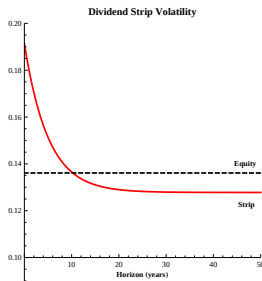
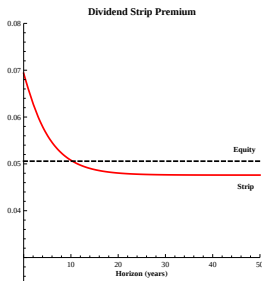
- The model-implied labor-share moves negatively with the VR's of dividends and positively with the distance between the VR's of wages and dividends.



- The model-implied labor-share moves positively with dividend growth.

Asset Pricing Results

- Premium and volatility of dividend strip returns feature negative slope:



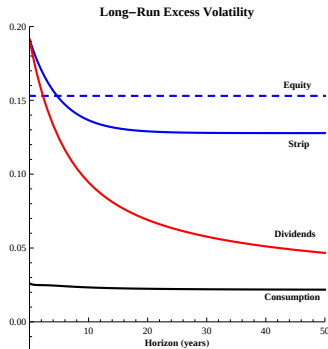
- The stronger income insurance, the larger the premium level, the steeper the negative slope.
- Model accurately matches the magnitude of the slope:

	data*	model
- average excess spot return of dividend strips up to 5-years over market excess return:	1.2%	1.1%
- average excess futures return of dividend strips up to 5-years over market return in excess of 10-years bond yield:	4.1%	3.4%

*van Binsbergen and Koijen (2016 JFE)

Asset Pricing Results

- Equity and dividend strips feature long-run excess volatility over fundamentals:



- Long-run excess volatility is increasing with ϕ :

$$\sigma_P^2(t, \infty) - \sigma_D^2(t, \infty) = \frac{(1-2\psi)\sigma_\mu^2}{\lambda_\mu^2 \psi^2} + \frac{d_z^2 \sigma_z^2}{\psi^2}.$$

- Long-run risk models need $\psi < 1/2$.

Asset Pricing Results

- Model calibration reconciles the term-structures of both fundamentals and equity with standard asset pricing facts.
- Solution of the equity premium and risk-free rate puzzles under standard preferences:

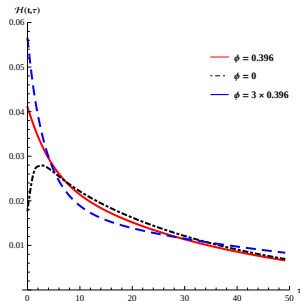
Data										
sample	r %	σ_r %	$\mu_P - r$ %	σ_P %	SR %	$\log P/D$	$\sigma_{\log P/D}$ %			
1931-2009	0.60	3.00	6.20	19.8	31.3	3.38	45.0			
1947-2009	1.00	2.70	6.30	17.6	35.8	3.47	42.9			
Model – Baseline calibration										
γ	ψ	β %	r %	σ_r %	$\mu_P - r$ %	σ_P %	SR %	$\log P/D$	$\sigma_{\log P/D}$ %	strip premium slope
10	1.5	3.5	0.86	2.80	5.06	13.6	37.2	3.51	41.3	< 0
Model – Walrasian benchmark (no income insurance: $h = 1 \Rightarrow \phi = 0$)										
γ	ψ	β %	r %	σ_r %	$\mu_P - r$ %	σ_P %	SR %	$\log P/D$	$\sigma_{\log P/D}$ %	strip premium slope
10	1.5	3.5	4.68	1.24	0.24	2.0	11.8	3.54	1.00	≈ 0

- Danthine Donaldson (2002) result is preserved once term-structures are taken into account.

Asset Pricing Results

- Whence the equity premium comes from?
- Horizon decomposition of the equity premium:

$$\mathcal{H}(\tau) = \frac{\Pi(\tau)}{\mu_P - r} \quad \text{with} \quad \mu_P - r = \int_0^\infty \Pi(\tau) d\tau$$

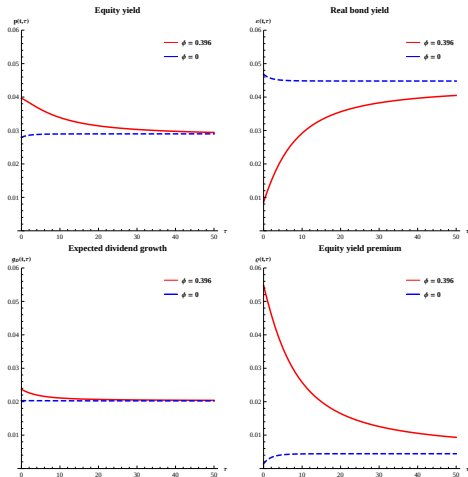


- The timing of macroeconomic risk suggests markets compensate risk at short horizons.
- Income insurance provides a rationale.

Asset Pricing Results

- Implied premium on the dividend strip yield to maturity:

$$\varrho(\tau) = p(\tau) - b(\tau) + g_D(\tau).$$



- Slopes are consistent with the empirical evidence.

- ① Motivation & Aim
- ② Model Intuition
- ③ Empirical Support
- ④ Model
- ⑤ Main Results
- ⑥ Model Extension**

Model Extension

Slope of equity premia is time-varying: slightly positive in good times and strongly negative in bad times.

van Binsbergen et al. (2013 JFE), Aït-Sahalia Karaman Mancini (2015), van Binsbergen Koijen (2016 JFE)

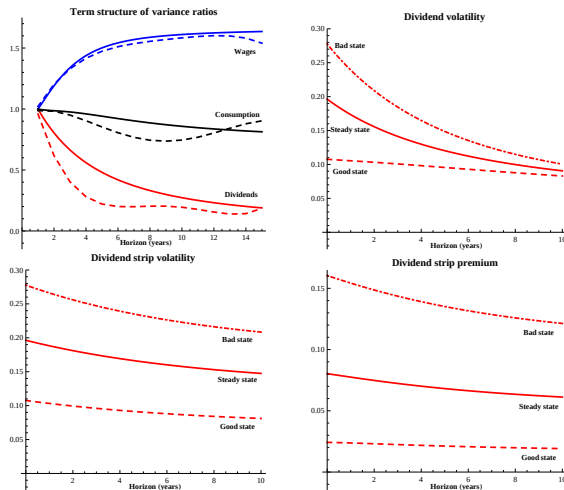
Model captures such a dynamics under business-cycle uncertainty:

$$\begin{aligned}\log Z_t = z_t &= \bar{z} - \hat{z}_t \\ d\hat{z}_t &= \lambda_z(\bar{z} - \hat{z}_t)dt + \sigma_z \sqrt{\hat{z}_t/\bar{z}} dB_{z,t}.\end{aligned}$$

- income insurance protects wages from short-run aggregate uncertainty
- counter-cyclical equity premium and Sharpe ratio
- LRR dominates SRR in good times (high z_t)
- SRR dominates LRR in bad times (low z_t)

Same calibration procedure preserves previous results and leads to pro-cyclical slope dynamics for dividend strip premia.

Model Extension



variance ratios (upper left), dividend volatility (upper right), dividend strip volatility (lower left), dividend strip premia (lower right).

Conclusion

- Empirical analysis:
 - income insurance explains the timing of wage and dividend risk,
 - income insurance is a first driver of dividend strip risk, premium and slope.
- Simple & parsimonious g.e. model with income insurance from shareholders to workers.
- Dynamics of wages and dividends subject to distributional risk.
- Downward sloping term-structures of equity risk and premia together with high equity premium, excess volatility, low and smooth risk-free rate.
- Additional asset pricing implications: counter-cyclical Sharpe ratio, increasing real bond yields, dynamic slope of dividend strip premia, excess return predictability. Also cash-flows duration premium and value premium (other paper).

Additional Material

Similarly to Lettau Watcher (2007 JF, 2011 JFE):

- single-firm characterized by deterministic (hump-shaped) lifetime dividend share process $s(T)$, $T \in (0, T^{\max})$
- dividend-shares determine the stable distribution of firms over time
- value firms are shorter-horizon equity than growth firms:
i.e. value firms' cash-flows are weighted more toward the present (large T)
and growth firms' cash-flows more toward the future (small T)

Lettau Watcher exogenously specify SDF to price SRR and generate value premium.

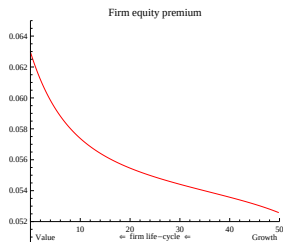
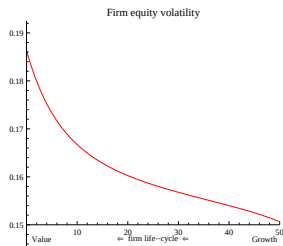
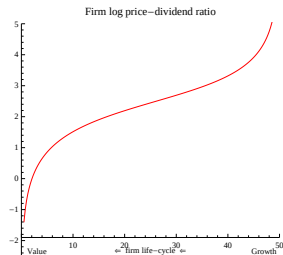
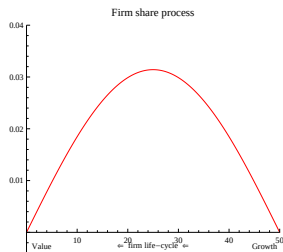
Here, SRR is endogenously priced in equilibrium and enhanced by income insurance:

- $\phi > 0 \Rightarrow$ value premium
- $\phi = 0 \Rightarrow$ growth premium

Income insurance rationalizes the intuition by Lettau Watcher.

close connection between labor rigidity and both term-structure and cross-sectional properties of equity (Marfè 2015).

Additional Material



share process/distribution (upper left), log price-dividend ratio (upper right),
return volatility (lower left), cross-sectional premium (lower right).

Additional Material

- Labor-share predicts the value premium:

horizon	1	2	3	5	7
W/Y	1.22**	1.61**	1.90***	1.68***	1.17***
NW t-stat	(2.05)	(2.44)	(2.89)	(3.26)	(3.13)
adj-R ²	0.01	0.08	0.18	0.21	0.15

- Labor-share predicts the duration premium:

horizon	1	2	3	5	7
W/Y	1.03	1.72**	2.11**	2.53***	2.06***
NW t-stat	(1.18)	(2.06)	(2.52)	(3.08)	(2.78)
adj-R ²	-0.01	0.03	0.09	0.20	0.18

- Value premium and duration premium are highly correlated across horizons:

horizon	1	2	3	5	7
corr	0.62	0.76	0.79	0.79	0.77

- Labor-share does not predict the value premium orthogonalized wrt the duration premium:

horizon	1	2	3	5	7
W/Y	0.79*	0.68	0.79*	0.39	0.18
NW t-stat	(1.68)	(1.49)	(1.83)	(1.23)	(0.63)
adj-R ²	0.00	0.02	0.07	0.01	-0.01

Additional Material

